

Role of the Local Electric-Field in High-Field Conditioning of DC Electrodes: Numerical and Experimental Insights

Victoria M. Bjelland,^{*} William L. Millar, and Walter Wuensch[†]
Espl. des Particules 1, 1217 Genève, CERN

Morten Kildemo
Department of Physics, NTNU, Heogskoleringen 5
(Dated: February 2, 2026)

Conditioning, the progressive increase of voltage-holding through the application of fields in a controlled way, is an important and commonly used process to bring high-field and high-voltage devices up to their full operating parameters. Here, a study is presented on how conditioning can vary within a device, specifically, when there is a spatial variation in the surface electric field. What has been observed in high-field pulsed direct current electrodes and radio frequency structures is that locations that are exposed to higher field exhibit an increased tendency to breakdown, but this increase is counteracted by an increased conditioning rate. This interplay explains observed breakdown locations and provides important insights into the mechanisms that underlie the conditioning and breakdown processes. This study combines Monte Carlo simulations with results of experiments using pairs of high-field electrode with lateral varying electric field. Results are presented from a high-field pulsed DC system, where the position of each breakdown during conditioning was recorded by triangulation using a pair of cameras and results are compared to the Monte Carlo simulations.

I. INTRODUCTION

In many high-field and high-voltage devices, it is commonplace to gradually increase the operating voltage while monitoring for arcs, as opposed to immediately commencing operation at the design voltage. Doing so typically yields an increased propensity for establishing high electric fields without arcing. This procedure is known as conditioning, and is an essential step in the commissioning process of vacuum interrupters, and normal-conducting particle accelerator cavities and waveguide components [1–3].

The investigation of field-driven conditioning has both practical and theoretical benefits. Conditioning procedures are time-intensive processes and typically carry an inherent risk of component damage. An improved understanding of the process can aid in identifying material, manufacturing, and treatment choices that can mitigate the need for conditioning, reduce the associated risks, and facilitate the optimization of existing procedures. A prominent example is the high-gradient radio-frequency (RF) cavities developed for applications ranging from cancer treatment to future collider concepts, i.e. flash therapy and CLIC [4–6]. Cavities of this kind operate at surface fields up to ~ 220 MV/m, and generally required substantial conditioning periods [5–8].

To shed light on the mechanisms that underlie conditioning, a Monte Carlo simulation was previously developed at CERN [9]. This report describes the design of two electrodes based on simulations performed using this tool, followed by a dedicated experiment that was carried

out in the CERN pulsed DC system. To investigate the field dependence of conditioning, a tapered electrode design was employed to establish an inhomogeneous surface electric field. The electrodes were conditioned to a peak surface electric field of approximately 80 MV/m and the positions of the breakdowns were recorded on a pulse-to-pulse basis. The spatial variation in the surface electric field combined with precise temporal and spatial breakdown data serves two main purposes. Firstly, the results shed information on the mechanisms associated with conditioning and breakdown by clearly illustrating the interplay between the applied field producing a conditioning effect and driving breakdown, and provide information about the relative strength of each phenomenon. Secondly, the results serve as an important dataset against which models of conditioning and breakdown, such as the one presented herein, may be benchmarked.

II. FEATURES OF HIGH-FIELD CONDITIONING

To date, numerous experiments have been conducted to explore the mechanisms associated with high-field vacuum arcs and conditioning. Parameters that have been studied include the device geometry, material, operating frequency (including DC), the gaps between electrode surfaces, and the high-field pulse length and repetition rate [10–15]. Recently, DC and RF tests have also been performed at cryogenic temperatures and this concept now forms the basis of a proposed future collider concept [16–20], while in the context of existing facilities attempts have been made to discern the effects of radiation exposure on high-field performance [21, 22].

A comprehensive set of models that describe the different experimental signatures, dependencies and stages

^{*} victoria.bjelland@cern.ch; Also at NTNU

[†] walter.wuensch@cern.ch

of breakdown has been developed. The breakdown process first involves material dislocation dynamics driven by the stress from the applied surface electric field, followed by the formation of surface features, their evolution driven by surface dynamics and finally heating, electron emission and field driven neutral atom emission and finally plasma formation. A review of the current status of the understanding of the breakdown process is given in [23]. An improved understanding of conditioning will contribute to further progress in understanding breakdown.

A. Conditioning Procedures

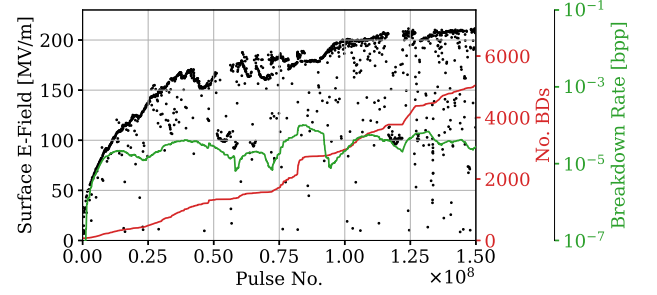
For the purposes of this work, conditioning is taken to mean the increased propensity for establishing high-fields without arcing or increases in internal pressure that prohibit operation in normal-conducting vacuum devices. In the early stages of operation some high-field devices, particularly RF cavities, are commonly limited by multipacting and elevated pressure levels, as opposed to arcing exclusively [24–27]. In one case, an algorithm was developed specifically to condition cavities using the pressure level as the process variable [28]. However, following this period the rate at which the operating field can be increased is often predominantly limited by the occurrence of vacuum arcs. Historically, conditioning procedures were often carried out based on anecdotal or empirical evidence, as opposed to a precise scientific understanding of the process. However, automated algorithmic approaches to conditioning are now commonplace, and offer a more consistent and systematic approach to component testing and commissioning [2, 28–31].

Generally, conditioning procedures, both manual and algorithmic, are based on the breakdown rate, i.e., the number of vacuum arcs that have occurred within a chosen temporal window, or equivalently, number of pulses. The pulse width and field level are then adjusted in order to maintain a target breakdown rate until the final operating parameters are reached. Practically speaking, this equates to the operating field being progressively increased while monitoring for arcs, in order to maintain a permissible rate of arcing. However, additional action is also occasionally required to respond dynamically in the event of, e.g., an elevated arc rate or otherwise anomalous behaviour.

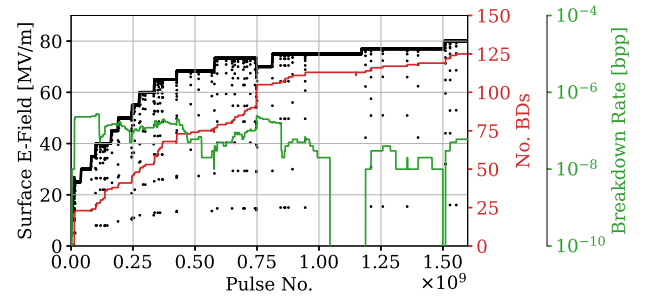
B. Characteristics of Conditioning

During conditioning procedures, if a constant breakdown rate is maintained, the rate at which the field may be increased typically decreases, resulting in the emergence of asymptotic behaviour. To illustrate this behaviour, the conditioning of an X-band accelerating structure and a DC electrode are shown in Fig 1 (a). The approach of progressive field ramping has been shown to

reliably facilitate operation at high fields with reduced probability of arcing, and a major theoretical objective of this work is to gain insight into the mechanism that produces this behavior.



(a) The conditioning curve of a 12 GHz copper RF cavity (T24 PSI N1 [15]).



(b) The conditioning curve of a copper electrode (NAME?) that was tested in the LES at CERN.

Figure 1: Conditioning curves of a 12 GHz copper RF cavity (T24 PSI N1 [15]) that was tested in CERN's high-gradient test facility (a) and a copper DC electrode (NAME?) that was tested in the LES with a gap size of 60 μm (b), illustrating a typical asymptotic increase in applied field. Note that the x-axes differ in scale by an order of magnitude. **Actually, this is just the sloped electrode, so it definitely shouldn't be placed here. Do we have another electrode I can reference for tuning the model??? Another test will probably have more breakdowns since the surface isn't sloped.**

Long-term tests performed on X-band (~ 12 GHz) RF cavities have shown that the conditioning of similar devices proceeds most comparably in terms of cumulative high-field pulses, as opposed to cumulative the number of breakdowns [32]. This result has led to the suggestion that the conditioning effect, i.e. the increased field-holding capability, is produced by the application of high fields and is not solely a consequence of the arcs. Therefore, DC electrodes are conditioning in a stair-case fashion, where the applied electric field is increased more gently, see Fig. 1 (b). The conditioning approach is explained in more details in Section III. This result forms the basis for much of the work conducted in this manuscript and is discussed further in Section IV.

C. Characteristics of Vacuum Arcs

As they constitute one of the main limitations on stable high-field operation, significant effort has gone into improving the understanding of vacuum arcs, as well as predicting and mitigating their occurrence [33]. In various RF and DC high-field tests it has been shown that if an arc occurs following a quiescent period it is common for one or several more arcs to occur in quick succession, and that these groups of events are likely to occur in close proximity [34]. This observation has led to the grouping of arcs into primary or trigger events, which occur stochastically, and secondary events, which are thought to be a consequence of the former arc [15, 34, 35].

To date, several quantities have been studied as predictors of device performance. Examples include the group velocity in RF devices [36, 37], the ratio of the power flow to the iris size in accelerator cavities [38], and the so-called modified Poynting vector, S_c [39]. Although RF cavity studies have shown that, for a given breakdown rate, similar peak magnetic fields are obtained for different designs [12], post-test optical examinations have shown that the arcs predominantly occur where the surface electric field is highest [40]. An example is shown in Fig. 2, where the surface of an X-band RF cavity that operates in the TM_{110} mode is presented with the surface electric field distribution and the arc locations overlaid. In this mode, the regions where S_c , the surface electric field, and the surface magnetic field are well separated, and a post-test optical examination showed that the majority of breakdown sites were present where the electric field is highest [40].

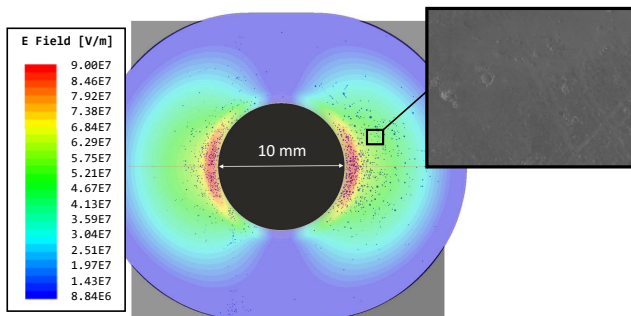


Figure 2: Electric field distribution on the second cell of the CLIC crab cavity with the breakdown locations superimposed (shown by the dark blue spots). Figure taken from a previous publication [9]. The test procedure and post-test analysis are reported in full elsewhere [40].

More broadly, a comprehensive analysis of long-term RF cavity tests found the breakdown rate to scale with the electric field E , the thirtieth power, or E^{30} [39]. More recent tests have shown similar power law relations, with reported exponents ranging from 28 to 35 [15, 41]. However, this scaling emerges globally and weaker depen-

dences have been reported across regions of inhomogeneous field in a given device [40], as in Fig. 2. In pulsed RF devices, the breakdown rate is commonly quoted as scaling with the pulse length, τ , to the fifth power, τ^5 [32, 39]. However, recent attempts to measure this dependency have yielded lower exponents, and the interpretation of such measurements is occasionally complicated by the use of non-rectangular pulse shapes [15, 42, 43]. In a previous study, the potential influence of the order in which the measurements are conducted was also noted [15].

D. Motivation

Generally, arcs serve as the feedback mechanism during conditioning procedures, with the occurrence of an arc being taken as an indicator that the field level is sufficiently high to produce a non-negligible conditioning effect. During such procedures, the field can gradually be increased while maintaining a constant rate of arcing. Conversely, if the applied field is held constant, measurements have shown that the arc rate gradually decreases. These observations suggest that the application of high electric fields increases the likelihood of arcing, and that the arc rate and/or conditioned state of the surface continuously evolve over the course of a given measurement. The objective of this work is to investigate the interplay between these two effects, and to provide information on the relative strength of each mechanism.

In multicell travelling wave (TW) RF cavity tests, localisation of arc sites to a specific cell is often accomplished in situ by employing a time-of-propagation based method [44]. However, localisation to a specific region on the surface is generally not possible without a post-test visual examination. Consequently, the exact spatial and temporal evolution of the arc distribution on the surface is lost. In some conditioning studies performed using DC electrodes, precise real-time breakdown localisation is possible, though only samples that support homogeneous electric fields have thus far been tested [45]. In this study, this limitation is addressed by testing a pair of electrodes that establish an inhomogeneous surface electric field while recording the arc locations and the order in which they occur. The electrodes were designed with the aid of a Monte Carlo model of conditioning [9]. The target of the optimisation process to select a geometry that, after conditioning, would have a gradual decrease in the arc density across its surface.

Therefore, this dataset represents, to the authors' knowledge, the first time that the positions of arcs and the order in which they were accrued have been recorded simultaneously in a pulsed device that operates with an inhomogeneous surface electric field. The results also serves as a means of validating the model with which the electrodes were designed, thereby confirming its suitability for more pragmatic applications such as the optimisation of existing conditioning procedures.

Notably, at the High Energy Accelerator Research Organization (KEK) a camera was integrated into a high-power cavity test, allowing direct in-situ observation of breakdowns as they occurred [46]. However, this arrangement was operated in a continuous wave regime and focused on the mechanisms associated with breakdown triggers, as opposed to the evolution of the arc distribution and conditioned state of the surface that are the focus of this work.

III. EXPERIMENTAL SETUP (LES)

The experimental measurements were performed in a vacuum chamber equipped with a set of interchangeable electrodes. Because of the notably large electrode dimensions, the apparatus is widely known as the Large Electrode System (LES) [47–51]. Figure 3 a) shows a cross-section view of the setup. Inside, the two electrodes are separated using a precision-machined insulating spacer. The spacer rests on the cathode, while the anode sits atop the spacer. Two high voltage connectors (HVCs) are pin-type feedthroughs, specialized for giving and receiving power to and from the inside of a vacuum chamber. HVC_A and HVC_C connect to the anode and cathode, respectively. On the side, one of the two CMOS cameras is shown. The second is placed in a perpendicular position, at the backside of the shown chamber.

The chamber is evacuated using a single turbomolecular pump to achieve vacuum levels down to 10^{-9} mbar. Pressure is monitored along the vacuum line using a cold-cathode vacuum gauge, and recorded at a sampling rate of 3 Hz.

The electrodes typically comprise of a flat machined ($R_a < 2$ nm) metallic cathode and anode to ensure a spatially uniform surface electric field. Depending on the chosen spacer, the typical electrode gaps are 20, 40, 60, or 100 μm . The gap distance is verified by capacitance measurements, as the electrodes act as a parallel-plate capacitor. The exposed radii of the cathode and anode are 30 mm and 20 mm, respectively. These dimensions remove edge-related field enhancement, which would localize breakdown events at the electrode periphery due to the increased surface electric field [48].

The electrodes consist of oxygen free (OFE) Cu electrodes, which were heat treated in hydrogen atmosphere up to 800 °C to extend the conditioning time [52]. The anode was machined with a sloped edge, where the central region remains flat.

Therefore, the gap height $d(r)$ can be found as a function of the radius

$$d(r) = \begin{cases} 60 \mu\text{m}, & r \leq 6.5 \text{ mm} \\ 60 \mu\text{m} + (r - 6.5 \text{ mm}) \tan(\theta), & r > 6.5 \text{ mm} \end{cases} \quad (1)$$

where $\theta = \tan^{-1}(10 \mu\text{m}/23.5\text{mm})$. From here, the sur-

face electric field can be found as

$$E_{\text{surface}}(r) = \frac{V_{\text{applied}}}{d(r)}, \quad (2)$$

ensuring a uniform surface electric field when $r \leq 6.5$ mm and a radially decaying when $r > 6.5$ mm. The design details are given in section IV.

During pulse applications, short high-voltage pulses are applied to HVC_A provided by a pulsed high-voltage generator (EPULSUS-FPM1-10) [53], while HVC_C is grounded. The generator employs a Marx circuit configuration, charging ten capacitors in parallel before discharging them in series, hence the name Marx generator. The LES Marx generator can deliver pulses up to 10 kV and 50 A, with pulse widths ranging from 500 ns to 100 ms. The generator is powered by an external supply.

To condition, the voltage is incrementally increased from a chosen start voltage by 10 V every 100 000 pulse until a chosen target voltage is reached. Once reaching the target voltage, the system continues pulsing at this voltage level until a new target is given.

When a vacuum arc occur during a pulse, it is detected by a surge in the current demand from the Marx generator, prompting an immediate three-second pause before pulsing resumes. During recovery, the applied voltage is reduced to 20 % of its pre-breakdown value and then logarithmically ramped back up to the conditioning set-point [54]. If the breakdown rate (BDR), the number of breakdowns per pulse, exceeds a preset threshold (typically 10^{-5} bds/pulse), the system actively lowers the applied voltage until the BDR is met [54]. A collection of conditioning studies using this experimental setup can be found in literature [21, 22, 52, 54, 55].

Breakdowns are also accompanied by sharp pressure spikes and brief flashes of light lasting a few microseconds [56]. To access the light signals, the LES is equipped with four symmetric viewports. Two perpendicular CMOS camera are mounted on the system to enable spatial localization, as shown in Fig. 3 b).

A 100 μm wide slit aperture is added in front of each camera, enabling triangulation of the breakdown position. Two similar triangles (marked in blue) are created by drawing a line (marked as a red line in the figure) that goes through the breakdown position (x, y) , through the slit aperture and finally hitting the detector. The two measurable quantities, g_1 and g_2 , along with the system constants, a and b , are used to find the origin location with equations given in [47].

The cameras operate continuously having an exposure time of $\tau = 3$ s. Frames containing light signals are saved for analysis, while those without are discarded. The spatial positions are typically compared to post-processing images using optical images and microscopy techniques. More details on the detected signals are given elsewhere [47].

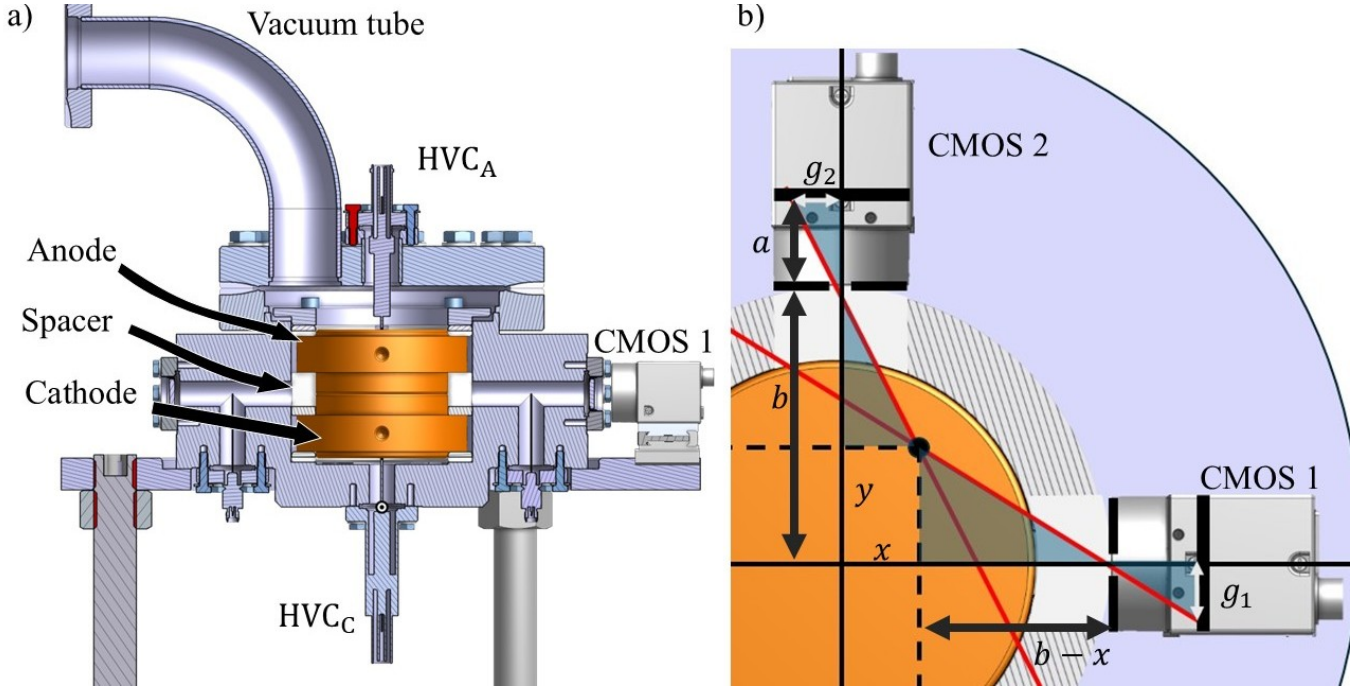


Figure 3: Cross-sectional views of the LES shown from the side (a) and from above (b). a) The electrodes are separated by an insulating spacer, creating a micrometer precision gap. Electrical contact to the anode and cathode are made via the HVC_A and HVC_C, respectively. Two CMOS cameras are attached to the four symmetrical viewports on the system. Only one shown here, as the other is hidden behind the system. b) Breakdown localization system. When a breakdown occurs, it creates a flash of light originating from (x, y) . Two red lines can be drawn going through the vacuum breakdown, through the apertures and hitting the detectors. This creates two similar triangles, which are marked in blue, where the original breakdown location can be calculated using known system constants a and b and the measured quantities g_1 and g_2 . The equations are given elsewhere [11].

IV. SIMULATION AND DESIGN OF A TAPERED ELECTRODE

Previously at CERN, a Monte Carlo-based model was developed to simulate the conditioning and operation of high-field devices on a pulse-to-pulse basis [9]. The model is deliberately constructed to be theory-agnostic, and relies on two main assumptions:

1. The application of an electric field produces a local conditioning effect, i.e., exposure to high electric fields increases the surface's propensity for establishing electric fields without arcing.
2. The probability of arcing scales strongly with the local electric field, with higher fields corresponding to an increased probability of arcing.

In addition to more pragmatic motivations, such as the optimisation of conditioning procedures, one of the key objectives was to investigate the interplay between these two counteracting effects, and the discrepancies in empirical scaling laws that have been reported. Although the model, and several preliminary studies conducted with it, have been discussed elsewhere [9], it has since undergone several modifications, and had not been applied

to DC devices. As it was employed in the design of the electrodes reported in this paper, and therefore forms the basis for much of the work reported herein, an overview is provided. Several peculiarities that pertain specifically to the simulation of copper electrodes operated in a pulsed DC regime are also discussed.

A. Overview of the Model

NOTE TO LEE, MAYBE CELL IS A MORE APPROPRIATE TERM AS THESE AREN'T TET MESH? The surface of the high-field device to be simulated is first meshed, yielding an array of n elements. Taking i as the element number, each element has the following attributes:

- a_i - The ratio of the mesh element area to the total simulated area.
- k_i - The ratio of the local electric field to the peak electric field for a given operating voltage or power level.

With this approach, the high-field conditioning of any arbitrary geometry and surface electric field distribution

is permitted. Additionally, the use of an inhomogeneous mesh permits the use of larger elements in regions where little activity, e.g., infrequent arcs or little conditioning, is expected to reduce the computational cost of the simulation.

Numerous experiments worldwide have shown that the voltage or power level that establishes a given breakdown increases asymptotically during conditioning procedures. Based on this observation, the model assumes that for a given pulse length and reference breakdown rate, P_{Ref} , there is a maximum, material-dependent surface electric field, E_L , in MV/m that a device can be conditioned to operate at. More succinctly, this assumption forbids the unphysical procedure of conditioning to arbitrarily high field levels or low breakdown rates.

To capture the conditioning effect, each surface element is also a conditioning state, $E_{S,i}$, which refers to the field level in MV/m to which the mesh element has been conditioned to operate at. Each element is also assigned a state enhancement factor, ψ_i , which is initially selected randomly from an appropriately tailored Gaussian distribution, and reselected each time the element accrues an arc. Depending on the applied, peak electric field, E_O , which also has units of MV/m, the conditioning state of the surface elements is increased on a pulse-to-pulse basis as

$$\Delta E_{S,i} = \gamma_i \cdot \frac{E_O \cdot k_i}{E_{S,i}} \cdot \left[1 - \frac{E_{S,i}}{E_{L,i}} \right], \quad (3)$$

where γ_i is a constant with units of V/m that allows the model to be fit to experimental data, and varies depending on the material, operating frequency, and pulse length. Generally, the majority of the breakdowns that are accrued during conditioning procedures occur during the preliminary stages, while increasing the applied field up to the target level [15, 32]. As the pulse length is typically fixed during this preliminary stage, the use of a fixed value for γ_i is sufficient for the studies reported in this work.

Although some variation is present, in well-conditioned copper RF devices the breakdown rate has typically been observed to increase with the applied electric field to the thirtieth power, i.e., $BDR \propto E_O^{30}$, [15, 39, 41], and investigating whether this scaling accurately reproduces the behaviour of other devices, particularly those constructed from different materials or with different operating frequencies, is another objective of the model. Based on this empirically derived scaling, the probability of a particular grid element arcing on a given pulse, $P_{BD,i}$, is expressed as

$$P_{BD,i} = a_i \cdot \left[\frac{E_O \cdot k_i}{E_{S,i} \cdot \psi_i} \right]^{30} \cdot P_G, \quad (4)$$

where P_G is a probability that is applied to the grid elements individually to obtain a global breakdown rate

that is comparable to P_{Ref} in the later stages of conditioning. To derive a suitable value for P_G , the probability of a pulse not resulting in an arc can be defined it is more convenient to define. Assuming the system is fully conditioned everywhere and all ψ_i values are equal to one,

$$1 - P_{Ref} = \prod_{i=1}^n (1 - P_G \cdot a_i \cdot k_i^{30}). \quad (5)$$

Although a solution can be obtained using iterative numerical methods, an approximation can also be reasonably obtained. Taking the natural logarithm of both sides of Eq. 5 yields

$$\ln(1 - P_{Ref}) = \sum_{i=1}^n \ln(1 - P_G \cdot a_i \cdot k_i^{30}). \quad (6)$$

However, $P_{Ref} \ll 1$, with typical values being less than 1×10^{-4} . Therefore, by taking the first term of the Taylor series, a reasonable first-order approximation may be obtained as

$$\ln(1 - P_{Ref}) \approx -P_{Ref}. \quad (7)$$

As $P_G \cdot a_i \cdot k_i^{30} < P_{Ref}$ for any scenario with more than one grid element, the same holds true for the right side of Eq. 6. Rearranging the terms then yields

$$P_G \approx \frac{P_{Ref}}{G}, \quad (8)$$

where G is a geometry factor, which corresponds the surface integral of the surface electric field to the thirtieth power, defined

$$G = \sum_{i=1}^n a_i \cdot k_i^{30}. \quad (9)$$

Consequently, Eq. 4 dictates that if all ψ_i values are equal to one, every surface element has been conditioned to its limit, $E_{L,i}$, and the device is operated at this level, the breakdown rate will be approximately equal to P_{Ref} . This ensures that the performance in simulation is similar to the experimental data from which $E_{L,i}$ and P_{Ref} were selected, independent of the mesh resolution. In low-field regions, the assumption that the surface has been fully conditioned is not necessarily correct. However, the probability of breakdown scales extremely strongly with the applied field, as per Eq. 4, and so the error that this effect introduces is small.

In several previous studies, an alternative approximation was used [9], which instead relied on the assumption that

$$1 - P_{Ref} \approx (1 - P_G)^G. \quad (10)$$

Rearranging this expression, P_G can then be defined as

$$P_G = 1 - (1 - P_{Ref})^{\left(\frac{1}{G}\right)} \quad (11)$$

This approximation emerges from taking the natural logarithm of the left hand side in Eq. 10, exponentiating the result, and, as $P_g \ll 1$, once again taking the first term of the Taylor expansion to show

$$\begin{aligned} (1 - P_G)^G &= \exp(\ln(1 - P_G) \cdot G) \\ &\approx \exp(-P_G \cdot G) \\ \therefore 1 - P_{Ref} &\approx \exp(-P_G \cdot G), \end{aligned} \quad (12)$$

and therefore, taking the natural log once again and employing the approximation in given in Eq. 7, the same approximation as Eq. 7 emerges as

$$-P_{Ref} \approx -P_G \cdot G. \quad (13)$$

The definition in Eq. 11 holds if the variation in $a_i \cdot k_i^{30}$ products is small, and therefore is suitable for devices where the mesh and electric field distribution are uniform, or in cases where regions of lower electric field are compensated by much larger mesh cells. However, the form given in Eq. 8 yields an improved result when larger cell-to-cell variations are present, and is therefore more broadly applicable. For this reason, the Eq. 8 definition was employed for the studies presented in this work.

Regarding the selection of P_{Ref} and E_L , to accurately reproduce a device's behaviour the values that are chosen should ideally be taken from late-stage measurements conducted using the pulse length that was used in the early stages of conditioning. Measurements at multiple pulse lengths are often conducted during more comprehensive high-field tests [15, 57], and so this aspect of the model is generally not problematic. Additionally, as γ_i is presumed to be constant for a given set of device parameters, only a single appropriate test is required to derive an appropriate value.

Additionally, the P_{Ref} and E_L values are somewhat interchangeable, provided the γ value has been appropriately selected so as to provide the correct dynamic behaviour. Hence, to tune the model, the approximate final performance of a device must be known a posteriori. Once the material properties are known and verified however, the model may be applied to other geometries and devices by simply rescaling P_{Ref} to account for any changes in the surface area and electric field distribution in the new design. Taking a trivial example, if two identical reference devices were to be simulated in parallel, the surface area, and therefore reference breakdown rate should also double for a given field level. More generally, the chosen reference breakdown rate is adjusted to provide a scaled reference breakdown rate, P_{Ref}^S , as

$$P_{Ref}^S = \frac{A^S \cdot G^S}{A \cdot G} \cdot P_{Ref}, \quad (14)$$

where A^S and A are the surface area of the device being simulated, and the device from which P_{Ref} was taken respectively. Similarly, G^S and G are the geometric factors, as per Eq. 9, of the device being simulated, and the reference device respectively. The probability applied to the grid elements of the new device, P_G^S , is then defined like in Eq. 8, as

$$P_G^S \approx \frac{P_{Ref}^S}{G^S}. \quad (15)$$

and hence, for the new device Eq. 4 becomes

$$P_{BD,i} = a_i \cdot \left[\frac{E_O \cdot k_i}{E_{S,i} \cdot \psi_i} \right]^{30} \cdot P_G^S. \quad (16)$$

B. Electrode Design

The large electrode system is equipped with CMOS cameras that facilitate real-time positioning of arcs on the surface of the electrode under tests and was covered in Section III. However, the usual flat electrodes that have been tested in this system are not good candidates for exploring the field dependence of conditioning.

To address this limitation, a sloped, frustum-like electrode that is compatible with the LES was developed in order to establish a linear decrease in the electric field in the radial direction. In this arrangement, the evolution of the distribution of arcs on the surface can then be directly monitored during operation, providing insight into the the field dependence of conditioning. Additionally, the design is still relatively straightforward to manufacture, and the LES as a whole benefits from a significantly reduced complexity and cost relative to RF cavity tests.

To provide a reasonable foundation for the design of an appropriate electrode, the model described in Section IV A was first tuned by eye to approximately reproduce a standard, flat electrode (**Mention gap size**) based on the results of previous high-field copper electrode tests conducted at CERN. A simulation of a conditioning of a copper electrode using the field ramp shown in Fig. 1 is shown in Fig. 4, and chosen model settings are summarised in Table I. The total number of breakdowns accrued in the simulation is lower than in the experimental case, as the anomalous cluster of breakdowns that occurred around 750 million pulses in Fig. 1 and necessitated a reduction in operating field during the experiment was not reproduced by the model.

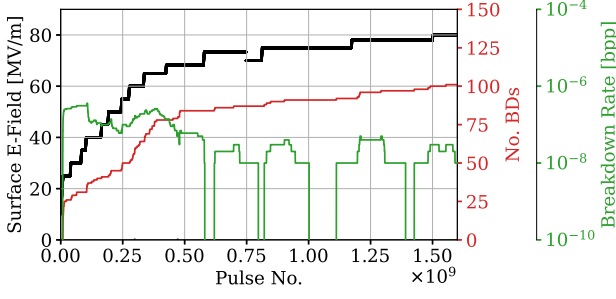


Figure 4: Simulated conditioning of a flat DC copper electrode. The model was tuned using on existing LES data to produce similar behaviour. The parameters employed are summarised in Table I. A total of 40 mesh cells were used.

Table I: The simulation settings used to approximately reproduce a copper cathode that was tested to a field level of 80 MV/m with a gap size of 60 μm . The μ and σ values are the mean and standard deviation of the Gaussian distribution from which the ψ_i values are selected respectively.

Quantity	Value	Unit
P_{Ref}	1E-7	-
Initial E_S	21E6	V/m
E_L	80E6	V/m
γ	0.25	V/m
μ	1	-
σ	0.1	-

A cross-section of the frustum electrode concept, including the relevant dimensions, is shown in Fig. 5. The LES system and existing spacers are designed to accommodate anodes with a radius r_0 of 20 mm and, as described in Section III, a gap size r_i of 60 μm is a standard and commonly used configuration for electrode tests. Accordingly, r_0 and h_i were fixed at 20 mm and 60 μm respectively.

Simulations were then performed using different combinations of r_i and h_2 . The objectives were to identify a combination of values that produce a clearly discernible decrease in the number of arcs per unit area in the outward radial direction, and to ensure that the flat central region remains sufficiently large to serve as a reference for normalisation. Given the azimuthal uniformity of the electrode, the surface was meshed using concentric bands. The electric field associated with each concentric band was calculated by evaluating the electrode separation at the radial midpoint of the band, effectively yielding a band-averaged field. An illustrative example using a relatively coarse mesh is shown in Figure 6.

In practice, some deviation from this idealised approximation arises due to field enhancement at electrode edges and the presence of fringe fields, particularly at the outer-

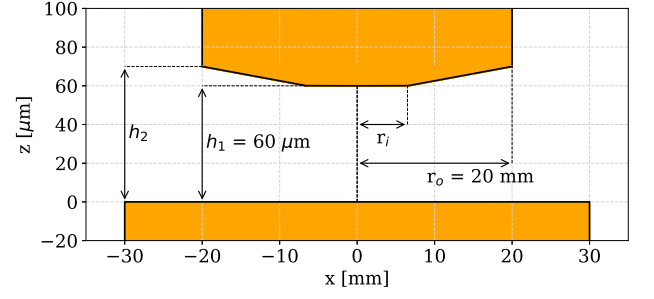


Figure 5: Cross-section of the tapered electrode geometry. Due to the small gap size, units of μm are used on the z-axis, while units of mm are used on the x-axis. Dimensions h_1 and r_0 are shown for completion, but were fixed at 60 μm and 20 mm respectively and not varied in simulation.

most band. However, arc-induced damage during high-field testing occurs predominantly on the cathode surface, and so the impact of these effects is mitigated by machining the relevant features on the anode rather than the cathode.

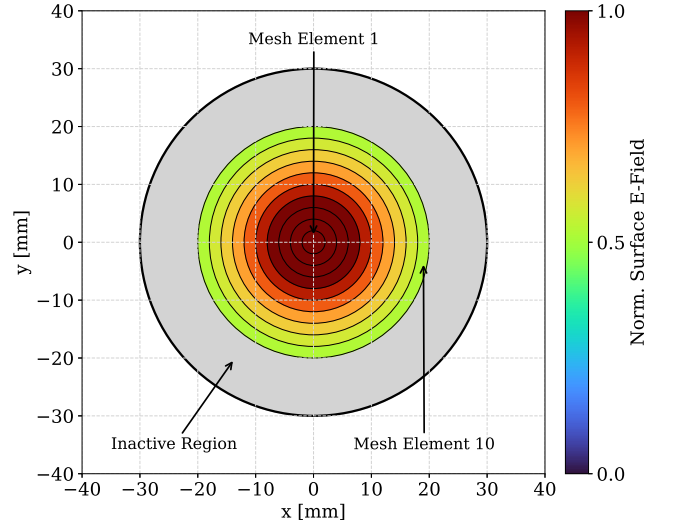


Figure 6: Top-down view of the simulated cathode surface. A total of ten mesh elements are employed in this example, as indicated by the coloured (non-grey) bands. The colour scale represents the electric field magnitude for each element. The field values shown correspond to an arbitrary arrangement in which $h_1 = 60 \mu\text{m}$, $h_2 = 120 \mu\text{m}$, $r_i = 8 \text{ mm}$, and $r_o = 20 \text{ mm}$. As the anode is smaller than the cathode the surface electric field in the grey region is negligible, and so this area is excluded from the simulation.

The results for three parameter combinations, including the chosen one, are shown in Figure 7. To obtain a more robust result, each parameter combination, i.e., anode geometry, was simulated 100 times and the results were averaged. The LES control system was included in the model and a typical conditioning procedure comprised of regular 5 MV/m field increments was employed. The chosen electrode dimensions, following the convention shown in Fig. 5, are summarised in Table II.

Table II: The chosen dimensions for the frustum electrode experiment, following the naming scheme shown in Figure 5. The slope angle is 0.0422° , and results in a field reduction 24 % at the edge of the active area on the cathode.

Quantity	Value	Unit
h_1	60	μm
h_2	70	μm
r_i	6.5	mm
r_o	20	mm

C. Predictions of the Model/Emergent Behaviour

In Fig. 8, the conditioning of three different mesh cells in the chosen design, each exposed to a different field, is plotted to illustrate the emergent behaviour that stems from the assumptions on which the model is founded, namely, that the conditioning effect is induced through applied pulses rather than solely through repeated breakdowns.

The average rate at which the field was increased differs for each cell, and hence, each was effectively subjected to a different conditioning procedure. After conditioning, the cells exposed to lower fields exhibit lower final breakdown densities, as expected. The ratios between the applied fields and the final breakdown densities for these elements are X:Y:Z and X:Y:Z, respectively. If the commonly reported E^{30} scaling were applied independently, a ratio of X:Y:Z between the breakdown densities would be intuitively expected.

In the present model, however, the E^{30} dependence is explicitly incorporated. Despite the difference in applied field, the difference in E_S that emerges is less pronounced, indicating that a significant conditioning effect is consistently produced in all elements. This result can therefore be understood as a consequence of the interplay between the applied field driving the probability of breakdown and producing the conditioning effect, and provides a natural explanation for the breakdown distribution observed in devices that operate with an inhomogeneous surface electric field distribution, such as the cavity shown in Fig. 2.

More notably, as indicated by the vertical lines in Figure 8, the different cells reach a given field level with

different breakdown densities. Testing for this behaviour is one of the key objectives of the experiment, as it has several implications. Firstly, this result supports the proposal that the conditioning effect is produced by applying high-field pulses, as well as by inducing breakdowns, and provides information on the relative strength of the mechanisms involved. From an operational perspective, it also suggests that devices can be conditioned with fewer breakdowns if less aggressive conditioning procedures are employed or, for example, if a sacrificial device or a digital twin were operated in parallel and used to regulate the conditioning. However, at the present time it is not known if such an approach would yield any benefit.

The model also grants access to quantities...mention Es or psi parameters here? Or is this too much detail. Probably too much for this paper...

V. EXPERIMENTAL RESULTS - VICTORIA

I think you should add a paragraph here to state what the conditioning procedure was i.e. what was algorithmic etc. At the very least, you should explicitly state that a "standard" conditioning procedure was used.

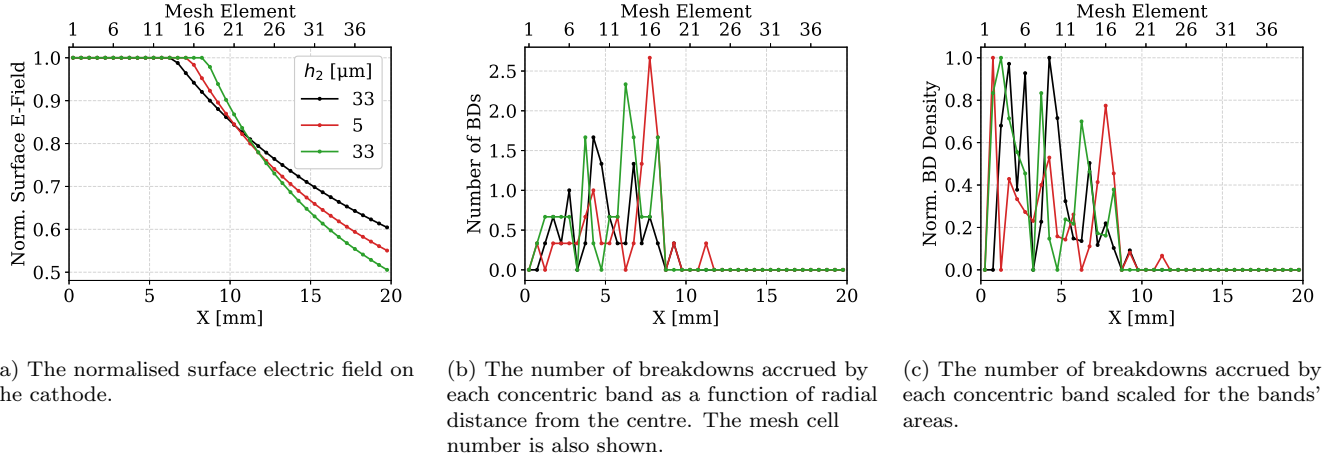


Figure 7: The surface electric field on the cathode (a), the total number of breakdowns accrued by each concentric band (b), i.e., mesh element, and the normalised total number of breakdowns per unit area for each concentric band (c) for simulations of three different anode geometries. **LEE TO REDO PLOTS, ARBITRARY DATA SHOWN**

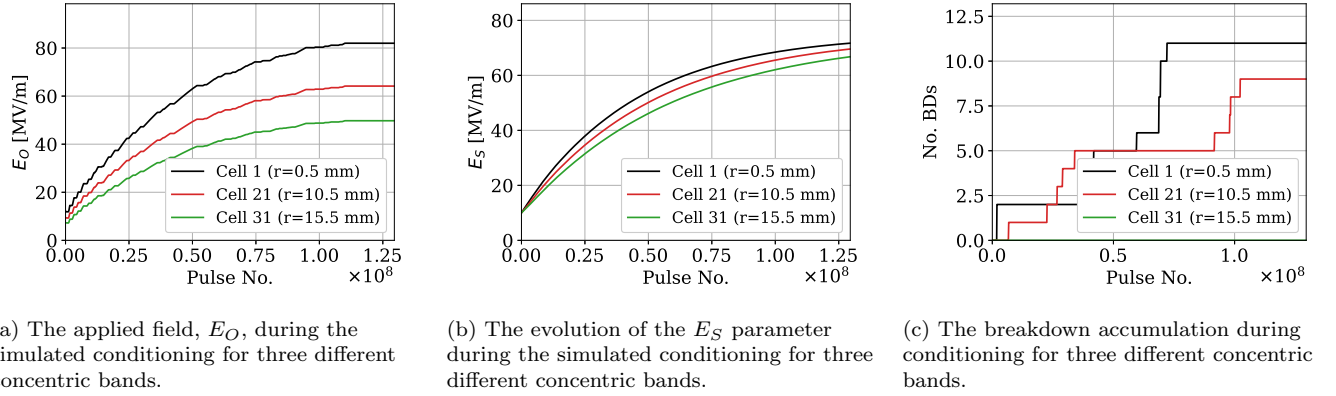
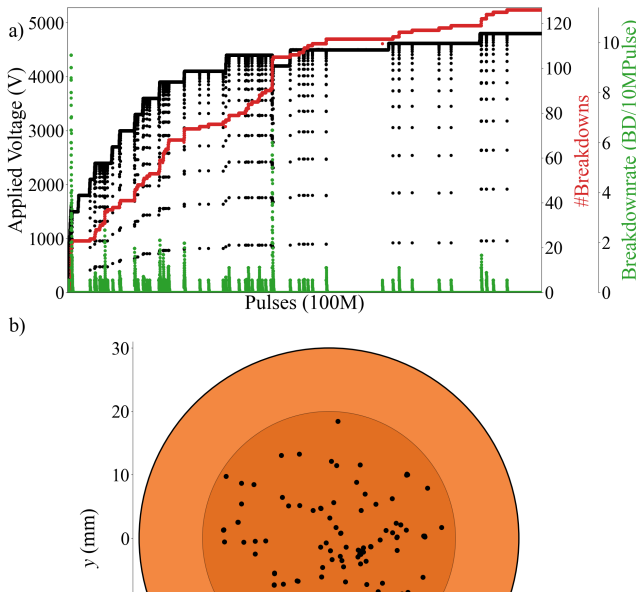


Figure 8: The evolution of E_O (a), E_S (b), and breakdown density (c) for three different concentric bands during the simulation conditioning. Also shown is the point at which they reached XX MV/m, illustrating that different conditioning scenarios produce different....The area of each element was.... **TO BE UPDATED BY LEE. We should show BD density and not Es maybe...**

A. Conditioning and spatial breakdown locations



Note from Lee - The way the electrode has been divided should be explicitly mentioned. So we should state the middle portion is the 6.5 mm radius and then the step sizes used when dividing it. Then state we normalise for area anyway. Maybe also state that we use three radial bands, since there aren't many breakdowns.

The electrodes were conditioned using a pulse length of 1 μs at a repetition rate of 1 kHz. The conditioning curve is shown in Fig. 9 a), where the black line represents the applied voltage over the electrodes (V_{applied}), the red line shows the cumulative number of breakdowns, and the green line corresponds to the BDR. The surface electric field is calculated from Eq. 2. When the center of the electrode reaches a surface electric field of E_{center} , the outer edge experiences a reduced field strength, given by

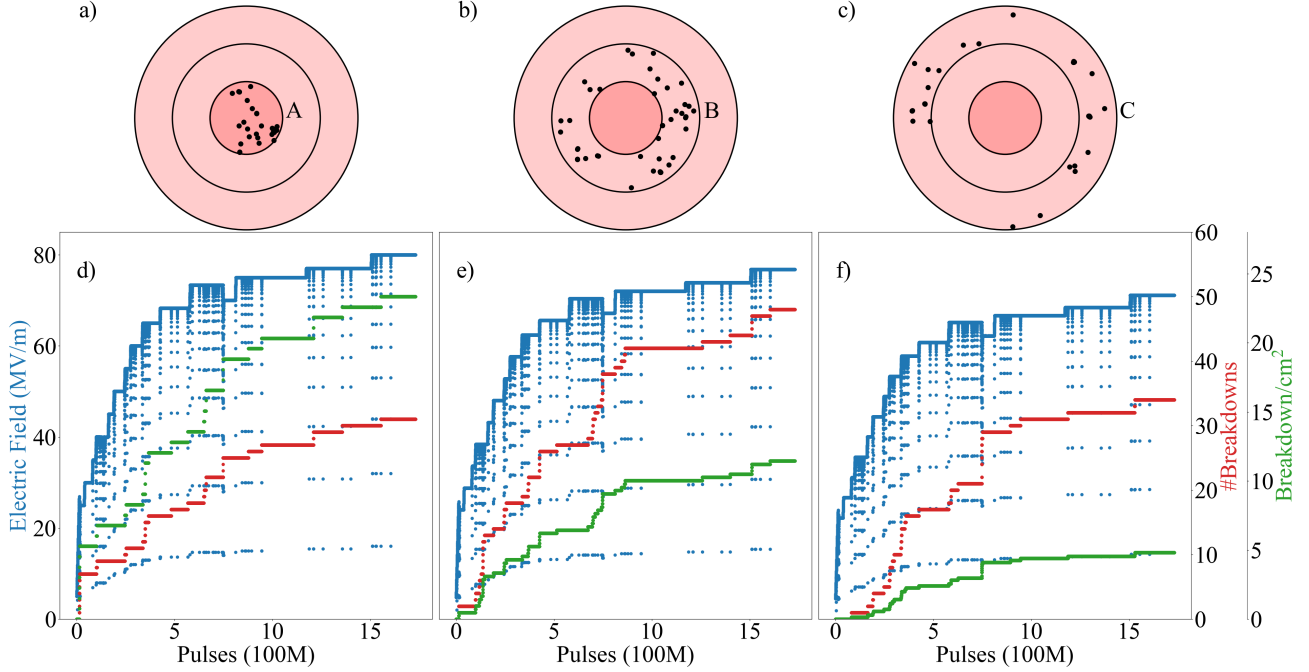


Figure 10: a-c) Spatially located breakdown positions, divided into sections A (inner), B (middle) and C (outer). d-f) Spatially divided conditioning where the blue lines are the average electric field of each region, the red lines are the accumulated breakdowns in each region and the green lines are the corresponding breakdown density of each region. **LEE- SHOULD WE HAVE ONE PLOT OF THE ELECTRODE ELSEWHERE IN THE PAPER TO LABEL THE REGIONS? Would it be better to have one plot for each parameter? i.e., having all the applied fields on one plot, all the bds on another... to make comparison easier. We can then draw a line where each region reaches xx MV/,m. TO BE DISCUSSED...**

$E_{\text{edge}} = 0.86E_{\text{center}}$. Since the center was conditioned to a surface electric field of 80 MV/m, the outermost edge was conditioned to a field of 68.8 MV/m.

Figure 9 b) shows the accumulated breakdown locations across the cathode electrode surface. The inner darker circle represents the surface electric field exposed area, consistent with the anode surface.

B. Spatial Division of Conditioning

To better understand the relationship between conditioning and breakdown behavior, the electrode surface is divided into three sections: A, B, and C. Spatially resolved breakdown events are categorized according to these sections, as shown in Figs. 10 a-c). Section A (inner) corresponds to the flat region of the anode, which exhibits a uniform surface electric field, while sections B (middle) and C (outer) are located on the sloped regions of the electrode.

Figures 10 d-f) present the conditioning data segmented by these spatial regions. In each figure, the blue line represents the spatially averaged surface electric field, the red line shows the cumulative number of breakdowns, and the green line indicates the spatial breakdown density (BDD), which is the number of breakdowns di-

vided by the active section, for the corresponding region. The maximum (spatially average) surface electric fields achieved in Sections A, B, and C were $E_{\text{inner}} = 80$ MV/m, $E_{\text{middle}} = 77$ MV/m and $E_{\text{outer}} = 71$ MV/m, respectively. The corresponding final breakdown densities were 24.0 BD/cm^2 , 12.2 BD/cm^2 , and 5.0 BD/cm^2 .

Figure 11 shows the breakdown number as a function of the radial distance r . The earliest breakdowns are predominantly concentrated in section A, followed shortly by events in section B and then section C. After approximately 30 breakdowns, the spatial distribution becomes more uniform, with breakdowns occurring across the entire electrode surface. This trend is also evident in Figure 10, where the breakdown number in each region increases throughout the conditioning process.

C. Comparison with Simulation

Due to the geometry of the anode, the surface electric field is inversely related to the radial distance as $E \propto 1/r$ when $r > 6.5$ mm. For the area inside the uniform surface field region, the BDD remains constant. Outside this

region, the BDD can be compared to r as

$$BDD = \begin{cases} \text{const}, & r \leq 6.5 \text{ mm} \\ \alpha(r - 6.5 \text{ mm})^{-\beta}, & r > 6.5 \text{ mm} \end{cases} \quad (17)$$

where α is a constant, removed by normalization, and β is the scaling factor. As previously addressed, β is reported to be between 28-35 for a global breakdown relation. However, in optical breakdown identification, it is observed to be lower. Figure 12 shows the experimental and simulated BDDs where the electrode is divided into 7 sections. The best fits of the experimental and simulated decays were found to be $\beta_e = 1.35$ and $\beta_s = 1.31$, respectively.

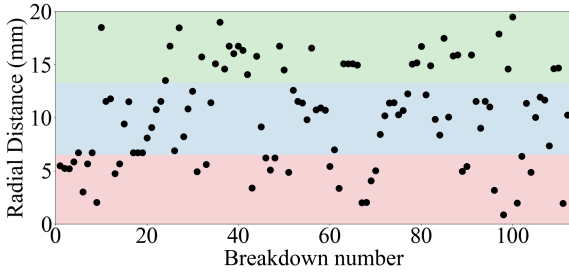


Figure 11: Breakdown number versus the distance from the center of the electrode to the breakdown.

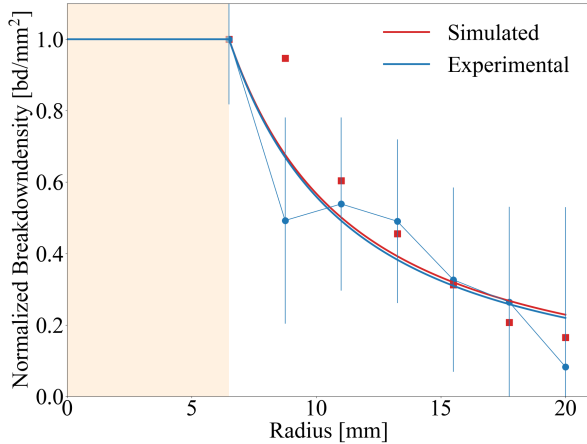


Figure 12: Polynomial fits to BDD over the radius of the electrode. The blue circles and red squares are the discrete experimental and simulated points, respectively. The blue and red lines are the fitted exponential decays. The scaling factor for the experimental data was found to be $\beta_e = 1.35$ and $\beta_s = 1.31$ for the simulated.

VI. DISCUSSION

We will now draw main insights from the experimental data and the corresponding Monte-Carlo simulations. Fig. 12 clearly shows the effect of field-dependent conditioning. This is because the spatial dependence of breakdown density on electric field is X-to-be-recalculated compared to 30 for the dependence of breakdown rate on applied voltage and from the agreement of the experimental data with the Monte Carlo simulation. LEE - IT WOULD BE GOOD IF WE MAKE HERE CONNECTIONS BACK TO MONTE CARLO ASSUMPTIONS AND PARAMETERS.

We can also see that breakdowns start in the highest field region, and progressively work their way out towards the lower field regions as the high field region becomes conditioned and can support higher fields. This too is exactly what is expected if a higher field drives the conditioning process more.

These results show that the total number of breakdowns that occur when conditioning to a specific field level depends on how the applied field is ramped. This can be seen in Fig. 10 by considering for example the 65 MV/m electric field reached in region C at the end of conditioning. BDDs of 5, 6.4 and 13.5 BD/cm² occurred reaching 71 MV/m in regions C, B and A respectively. This dependence occurs because the overall breakdown rate of the electrodes is used to control the applied voltage ramp-up speed in the conditioning algorithm, so the lower field regions are conditioned more slowly. This dependence also supports the previous observations that the number of pulses, rather than the number of breakdowns, condition the device, and extends the observation to local conditioning. [32]. VICTORIA -WE SHOULD ADD VISUAL GUIDANCE, POINTS OR A LINE, TO FIG. 8 TO MAKE IT EASIER TO SEE THIS POINT.

In more general terms, the agreement between Monte Carlo simulation and experiment show that there should be a physical manifestation of E_{state} , either on the electrode surface or in the sub-surface material structures. The results so far do not give a clear insight into what this might be. The dislocation dynamics model, which successfully predicts the dependency of breakdown rate on field and temperature, could fulfill such a mechanism. In this case the dislocation population in the sub-surface would evolve with for example mobile dislocations being pinned as conditioning progresses. Such a process would occur more quickly under the increased tensile stress on the surface due to increased applied field, as required for a physical manifestation of E_{state} .

I'VE COMMENTED OUT THE FOLLOWING TEXT BECAUSE I BELIEVE I'VE INCLUDED THE MAIN POINTS IN THE TEXT ABOVE. BUT PLEASE MAKE SURE THAT I HAVEN'T MISSED ANYTHING.

THE FOLLOWING PARAGRAPH CAN PROBABLY STAY BUT NEEDS WORK. SPECIFIC REFERENCES TO FIGURE NUMBERS IS NEEDED. THE

READER OF THIS PAPER WILL BE LOST BY THE REFERENCE TO THE EDGE ELECTRIC FIELD BEING HIGHEST FOR EXAMPLE.

In previous experiments comparing conditioning between soft and hard copper [55]. Soft copper, also known as heat treated copper, is known to need a longer conditioning time to reach the same field levels as hard copper. On the hard copper, the breakdowns are located on the outer electrode edge, where the surface electric field is at its highest. Since the rest of the surface is experiencing lower field levels, they are not as exposed to breakdowns. The soft copper on the other hand, requires more time to condition. This gives the field enhanced areas an opportunity to condition for longer time and become more resilient to breakdowns, which causes breakdown to occur at other low field regions on the surface, leading to the even distribution observed in the paper.

I'M NOT SURE ABOUT THIS PARAGRAPH. IT SEEMS TO BE COMPARING THE RELATIVE STATISTICAL ACCURACIES OF THE SIMULATION AND EXPERIMENT, BUT IT'S NOT CLEAR.

Both experimental and simulated data show the same general trend in the decay of breakdown density (BDD) as a function of radial distance r . However, the most intriguing aspect is the conditioning process itself. In experiments, the statistical data is limited by the number of breakdowns a single electrode can sustain. In contrast, simulations were run for over 1000 iterations, enabling more robust statistical analysis and a wider spatial distribution of breakdown events. This allows for a finer division of the surface area under investigation. While the experimental and simulated enhancement factors, β_e and $\beta_{e,s}$, were similar in value, the experimental estimates were subject to large errors and fluctuations. Interestingly, these fluctuations were less pronounced when the surface was divided into 5 to 12 sections instead of 7.

VII. SUMMARY AND CONCLUSIONS

We have demonstrated a new approach to testing conditioning and its effect on a local scale. By testing varying electric field electrodes, we investigated experimentally the relation between the conditioning effect of high field exposure with breakdowns. We compare with Monte Carlo simulations and show excellent correlation between the approaches.

From these data and corresponding Monte Carlo simulations we can see how the counteracting effects of the electric field to both drive breakdown and conditioning at the device level and how the effects resolve themselves within a device which has spatially varying electric fields.

In the future, the electrodes will be analyzed with electron microscopy to see if there is a spatial variation in the electrode surface, or in the sub-surface dislocation structure, to see if the mechanism of conditioning can be established. A change in the surface or material structure would allow us to assign a physical mechanism to the

E_{state} variable used in the Monte Carlo simulation. This in turn would give an important insight into the breakdown mechanism. It might also be possible to develop preparation procedures of electrode and radio frequency structures that shorten conditioning time. This could be done in a targeted way by trying to match the sub-surface structure modifications associated with conditioned areas with a simplified and less expensive procedure than high-field pulsing.

After this experiment was completed, the camera system was upgraded so that it can now spatially resolve emission sites (we have to see if we can reference this). In addition a complementary Monte Carlo simulation of conditioning has been published Cite Andreas, which ascribes the conditioning to changes in the population of field emitters. A future sloped electrode test would include recording of the evolution of the spatial distribution of field emission sites in addition to the locations of the breakdown sites. This could help to differentiate between proposed conditioning mechanisms.

In addition we should run the experiment much longer to see if the conditioning at the outer edges finally catches up with the center. In the case where the conditioning state of the surface is uniform the polynomial dependence fit shown in Fig.12 should return to the E^{30} whole-device value.

- [1] S. Kulkarni, D. Canat, C. Arnoux, and N. Kumar, in *Proc. of the 30th Int. Symp. on Discharges and Electrical Insulation in Vacuum (ISDEIV)* (2023) pp. 45–48.
- [2] L. S. Cowie, D. J. Scott, G. Burt, and W. L. Millar, in *Proc. of IPAC'18* (JACoW Publishing, Geneva, Switzerland, 2018) pp. 3852–3855.
- [3] N. Catalán Lasheras, H. Damerau, R. L. Gerard, A. Grudiev, T. Lucas, G. McMonagle, J. Paszkiewicz, S. Pitman, A. Solodko, I. Syratchev, A. Vnuchenko, M. Volpi, B. Woolley, W. Wuensch, and V. del Pozo Romano, in *Proc. of IPAC'18* (2018) p. WEPMF074, cERN-ACC-2018-156, CLIC-Note-1149.
- [4] R. Schulte, C. Johnstone, S. Boucher, E. Esarey, C. G. R. Geddes, M. Kravchenko, S. Kutsaev, B. W. Loo, F. Méot, B. Mustapha, K. Nakamura, E. A. Nanni, L. Obst-Huebl, S. E. Sampayan, C. B. Schroeder, K. Sheng, A. M. Snijders, E. Snively, S. G. Tantawi, and J. Van Tilborg, *Applied Sciences* **13**, 10.3390/app13085021 (2023).
- [5] M. Aicheler, P. Burrows, M. Draper, T. Garvey, P. Lebrun, K. Peach, N. Phinney, H. Schmickler, D. Schulte, and N. Toge, *A Multi-TeV Linear Collider Based on CLIC Technology*, CERN Yellow Reports: Monographs (CERN, Geneva, 2012).
- [6] O. Brunner, P. Burrows, S. Calatroni, N. Catalan, R. Corsini, G. D'Auria, S. Doeber, F.-G. Angeles, A. Grudiev, A. Latina, T. Lefevre, G. McMonagle, J. Osborne, Y. Papaphilippou, A. Robson, C. Rossi, R. Ruber, D. Schulte, S. Stapnes, and W. Wuensch, *The CLIC project*, Tech. Rep. (CERN, Geneva, 2022).
- [7] P. Martinez-Reviriego, N. Fuster-Martínez, D. Esperante, M. Boronat, B. Gimeno, C. Blanch, D. González-Iglesias, P. Martín-Luna, E. Martínez, A. Menendez, L. Pedraza, J. Fernández, J. Fuster, A. Grudiev, N. C. Lasheras, and W. Wuensch, *Nuclear Engineering and Technology* **57**, 103164 (2025).
- [8] B. Woolley, G. Burt, A. Dexter, R. Peacock, L. Millar, N. Catalan Lasheras, A. Degiovanni, A. Grudiev, G. McMonagle, i. Syratchev, W. Wuensch, E. Rodriguez Castro, and J. Giner Navarro, *Physical Review Accelerators and Beams* **23** (2020).
- [9] W. Millar, G. Burt, and W. Wuensch, in *Proc. LINAC'22*, International Linear Accelerator Conference No. 31 (JACoW Publishing, Geneva, Switzerland, 2022) pp. 283–286.
- [10] H. H. Braun, S. Döbert, I. Wilson, and W. Wuensch, *Phys. Rev. Lett.* **90**, 224801 (2003).
- [11] I. Profatlova, W. Wuensch, and S. Calatroni, *Behaviour of copper during initial conditioning in the pulse DC system*, Tech. Rep. (CERN, Geneva, 2019).
- [12] V. Dolgashev, S. Tantawi, Y. Higashi, and B. Spataro, *Applied Physics Letters* **97**, 171501 (2010).
- [13] W. Wuensch, C. Achard, S. Dobert, H. Braun, I. Syratchev, M. Taborelli, and I. Wilson, in *Proceedings of the 2003 Particle Accelerator Conference*, Vol. 1 (2003) pp. 495–497 Vol.1.
- [14] T. Lucas, M. Boland, N. Catalán Lasheras, P. Giansiracusa, A. Grudiev, T. Lefèvre, G. McMonagle, J. Paszkiewicz, R. Rassool, C. Serpico, I. Syratchev, A. Vnuchenko, M. Volpi, B. Woolley, W. Wuensch, R. Zennaro, and V. del Pozo Romano, in *Proc. of IPAC'18* (JACoW Publishing, Geneva, Switzerland, 2018) pp. 4548–4551.
- [15] W. L. Millar, A. Grudiev, W. Wuensch, N. C. Lasheras, G. McMonagle, R. Zennaro, P. Craievich, M. Bopp, T. G. Lucas, M. Volpi, J. Paszkiewicz, A. Edwards, R. Wegner, H. Bursali, B. Woolley, A. Magazinik, I. Syratchev, A. Vnuchenko, S. Pitman, V. d. P. Romano, D. B. n. Caballero, and G. Burt, *IEEE Transactions on Nuclear Science* **70**, 1 (2023).
- [16] M. Jacewicz, J. Eriksson, R. Ruber, S. Calatroni, I. Profatlova, and W. Wuensch, *Phys. Rev. Appl.* **14**, 061002 (2020).
- [17] M. Jacewicz, I. Profatlova, Y. Ashkenazy, S. Calatroni, I. Popov, P. Szaniawski, and W. Wuensch, *Journal of Applied Physics* **137**, 193302 (2025).
- [18] M. Nasr, E. Nanni, M. Breidenbach, S. Weathersby, M. Oriunno, and S. Tantawi, *Phys. Rev. Accel. Beams* **24**, 093201 (2021).
- [19] M. Schneider, A. Diego, A. Krasnykh, A. Dhar, E. Nanni, N. Matavalan, P. Carriere, R. Berry, and R. Agustsson, *JACoW IPAC2024*, MOPR29 (2024).
- [20] C. Vernieri, E. A. Nanni, S. Dasu, M. E. Peskin, T. Barklow, R. Bartoldus, P. C. Bhat, K. Black, J. E. Brau, M. Breidenbach, B. Bullard, N. Craig, D. Denisov, L. Gray, P. C. Harris, M. Kagan, Z. Liu, P. Meade, N. Majernik, A. Mohammadi, S. Nagaitsev, D. Ntounis, I. Ojalvo, K. Pachal, C. Paus, J. Rosenzweig, C. Schroeder, A. G. Schwartzman, E. Simakov, M. Swiatkowski, B. Spataro, J. Strube, S. Dong, S. Tantawi, L.-T. Wang, A. White, and G. W. Wilson, *Journal of Instrumentation* **18** (07), P07053.
- [21] C. Serafim, R. Peacock, S. Calatroni, F. Djurabekova, A. T. P. Fontenla, W. Wuensch, S. Sgobba, A. Grudiev, A. Lombardi, E. Sargsyan, S. Ramberger, and G. Belodi, *Frontiers in Physics* **Volume 12 - 2024**, 10.3389/fphy.2024.1308455 (2024).
- [22] C. Serafim, S. Calatroni, F. Djurabekova, R. Peacock, V. Bjelland, A. T. Perez-Fontenla, W. Wuensch, A. Grudiev, S. Sgobba, A. Lombardi, and E. Sargsyan, *Phys. Rev. Accel. Beams* **28**, 013101 (2025).
- [23] W. Wuensch, S. Calatroni, F. Djurabekova, A. Kyritsakis, and Y. Ashkenazy, *Fundamental mechanisms of high-electric-field vacuum arc initiation* (2025), arXiv:2502.03967 [physics.app-ph].
- [24] A. K. Mitra, Z. Ang, S. Calic, P. Harmer, S. Koscielniak, R. E. Laxdal, W. R. Rawnsley, and R. Shanks, in *Proceedings of LINAC2012* (TRIUMF, Tel-Aviv, Israel, 2012) tRIUMF, 4004 Wesbrook Mall, Vancouver, BC, Canada, V6T 2A3.
- [25] S. Papadopoulos, F. Gerigk, J.-M. Giguët, J. Hansen, J. M. Balula, A. Michet, S. Ramberger, N. Thaus, and R. Wegner, in *Proceedings of the 28th Linear Accelerator Conference (LINAC 2016)* (JACoW Publishing, East Lansing, Michigan, USA, 2017) p. THPLR060, cC-BY-3.0 license.
- [26] B. Xiao, S. Belomestnykh, J. Brennan, J. Brutus, G. McIntyre, K. Mernick, C. Pai, K. Smith, T. Xin, A. Zaltsman, and V. Veshcherevich, *Physical Review Accelerators and Beams* **22**, 10.1103/physrevaccelbeams.22.030101 (2019).
- [27] O. Piquet, M. Baudrier, A.-C. Chauveau, M. Desmons, P. Hamel, and et al., in *Proceedings of IPAC 2022* (JA-

- CoW, 2022) p. TUPOTK003, contribution to IPAC 2022.
- [28] B. Woolley, *High Power X-band RF Test Stand Development and High Power Testing of the CLIC Crab Cavity*, Phd thesis, Lancaster University (2015).
- [29] S. Benedetti, M. Cerv, S. Magnoni, J. L. N. Quirante, and S. G. Soriano, in *Proceedings of LINAC'22 (31st International Linear Accelerator Conference)* (JACoW Publishing, Geneva, Switzerland, Liverpool, UK, 2022) pp. 154–157.
- [30] E. Trachanas, C. Baltador, L. Bellan, G. Fedel, A. Gaget, and et al., in *Proceedings of the 17th International Conference on Accelerator and Large Experimental Physics Control Systems (ICALEPCS 2023)* (JACoW Publishing, Venice, Italy, 2023) p. 5, contribution TUPDP080.
- [31] T. G. Lucas, J. Alex, C. Beard, A. Citterio, H.-H. Braun, P. Craievich, Z. Geng, R. Kalt, F. Loehl, M. Pedrozzi, J.-Y. Raguin, and R. Zennaro, *IEEE Transactions on Nuclear Science* **72**, 781 (2025).
- [32] A. Degiovanni, W. Wuensch, and J. Giner Navarro, *Phys. Rev. Accel. Beams* **19**, 032001 (2016).
- [33] C. Obermair, T. Cartier-Michaud, A. Apollonio, W. Millar, L. Felsberger, L. Fischl, H. S. Bovbjerg, D. Wollmann, W. Wuensch, N. Catalan-Lasheras, M. m. c. Boronat, F. Pernkopf, and G. Burt, *Phys. Rev. Accel. Beams* **25**, 104601 (2022).
- [34] W. Wuensch, A. Degiovanni, S. Calatroni, A. Korsbäck, F. Djurabekova, R. Rajamäki, and J. Giner-Navarro, *Phys. Rev. Accel. Beams* **20**, 011007 (2017).
- [35] A. D. Cahill, J. B. Rosenzweig, V. A. Dolgashev, S. G. Tantawi, and S. Weathersby, *Phys. Rev. Accel. Beams* **21**, 102002 (2018).
- [36] C. Adolphsen, W. Baumgartner, K. Jobe, F. L. Pimpec, R. Loewen, D. McCormick, M. Ross, T. Smith, J. Wang, and T. Higo, in *Proceedings of the 2001 Particle Accelerator Conference, Chicago, IL, June 18–22, 2001* (2001) pp. 478–480, conf.Proc.C0106181, arXiv:physics/0108042 [physics.acc-ph].
- [37] C. Adolphsen, in *Proceedings of the 21st Particle Accelerator Conference (PAC 2005)* (JACoW, Knoxville, TN, USA, 2005) p. 204, sLAC-PUB-11224.
- [38] W. Wuensch, *The Scaling of the Traveling-Wave RF Breakdown Limit*, Tech. Rep. (CERN, Geneva, 2006).
- [39] A. Grudiev, S. Calatroni, and W. Wuensch, *Phys. Rev. ST Accel. Beams* **12**, 102001 (2009).
- [40] B. Woolley, G. Burt, A. C. Dexter, R. Peacock, W. L. Millar, N. Catalan Lasheras, A. Degiovanni, A. Grudiev, G. Mcmonagle, I. Syratchev, W. Wuensch, E. Rodriguez Castro, and J. Giner Navarro, *Phys. Rev. Accel. Beams* **23**, 122002 (2020).
- [41] P. Martinez-Reviriego, N. Fuster-Martínez, D. Esperante, M. Boronat, B. Gimeno, C. Blanch, D. González-Iglesias, P. Martín-Luna, E. Martínez, A. Menendez, L. Pedraza, J. Fernández, J. Fuster, A. Grudiev, N. C. Lasheras, and W. Wuensch, *Nucl. Eng. Tech.* **57**, 103164 (2025).
- [42] S. Matsumoto, T. Abe, Y. Higashi, T. Higo, and Y. Du, *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **657**, 160 (2011), x-Band Structures, Beam Dynamics and Sources Workshop (XB-10).
- [43] X. Wu, M. J. Boronat Arévalo, A. Castilla, N. Catalán Lasheras, A. V. Edwards, A. Grudiev, G. McMonagle, W. L. Millar, I. Syratchev, and W. Wuensch, *JACoW IPAC* **2021**, 1543 (2021).
- [44] R. Rajamäki, *Vacuum arc localization in CLIC prototype radio frequency accelerating structures*, Master's thesis, Aalto University, Helsinki (2016), supervisors: Walter Wuensch (CERN), Visa Koivunen (Aalto University).
- [45] I. Profatilova, X. Stragier, S. Calatroni, A. Kandrasyeu, E. Rodriguez Castro, and W. Wuensch, *Nucl. Instrum. Methods Phys. Res., A* **953**, 163079 (2020).
- [46] T. Abe, T. Kageyama, H. Sakai, Y. Takeuchi, and K. Yoshino, *Phys. Rev. Accel. Beams* **19**, 102001 (2016).
- [47] I. Profatilova, X. Stragier, S. Calatroni, A. Kandrasyeu, E. R. Castro, and W. Wuensch, *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **953**, 163079 (2020).
- [48] A. Saessalo, I. Profatilova, W. L. Millar, A. Kyritsakis, S. Calatroni, W. Wuensch, and F. Djurabekova, *Physical Review Accelerators and Beams* **23**, 10.1103/physrevaccelbeams.23.113101 (2020).
- [49] A. Saessalo, A. Kyritsakis, F. Djurabekova, I. Profatilova, J. Paszkiewicz, S. Calatroni, and W. Wuensch, *Physical Review Accelerators and Beams* **23**, 10.1103/physrevaccelbeams.23.023101 (2020).
- [50] R. Peacock, W. Wuensch, and G. Burt, in *2020 33rd International Vacuum Nanoelectronics Conference (IVNC)* (2020) pp. 1–2.
- [51] E. Z. Engelberg, J. Paszkiewicz, R. Peacock, S. Lachmann, Y. Ashkenazy, and W. Wuensch, *Phys. Rev. Accel. Beams* **23**, 123501 (2020).
- [52] I. Profatilova, W. Wuensch, and S. Calatroni, *Behaviour of copper during initial conditioning in the pulse DC system*, Tech. Rep. (CERN, Geneva, 2019).
- [53] F. Shi, P. Chen, S. Jiang, J. Zhuang, and J. Rao, *Electronics* **13**, 10.3390/electronics13010101 (2024).
- [54] R. Peacock, *Experimental Investigation of Vacuum Breakdown Triggering Mechanisms in a DC Electrode System*, Ph.D. thesis, Lancaster U. (2023).
- [55] A. Korsbäck, F. Djurabekova, L. M. Morales, I. Profatilova, E. R. Castro, W. Wuensch, S. Calatroni, and T. Ahlgren, *Phys. Rev. Accel. Beams* **23**, 033102 (2020).
- [56] J. W. Kovermann, *Comparative Studies of High-Gradient Rf and Dc Breakdowns*, Ph.D. thesis, Aachen, Tech. Hochsch. (2010).
- [57] X. Wu, J. Shi, H. Chen, J. Shao, T. Abe, T. Higo, S. Matsumoto, and W. Wuensch, *Phys. Rev. Accel. Beams* **20**, 052001 (2017).